

MHDRM Cross-Site Readiness Assessment

Prototype screening report for multimodal federated learning

Assessment Context

Project name	Scenario 1: AI Breast Cancer Detection Tool
MMFL use case	Development of an AI-enabled screening tool for early breast cancer detection.
Target outcome or label	Binary patient-level outcome: cancer / no cancer.
Participating sites	Hospital 1 (NY) Hospital 2 (MA) Hospital 3 (CT) Outpatient oncology center Cancer research institute
Target population	Adults at risk of breast cancer, with subpopulations grouped by age, demographics, and clinical risk factors.
Required modalities	Unstructured EHR data (e.g., clinical notes), Lab test results, Medical imaging (e.g., X-ray, CT, MRI)

Assessment Summary

Total score	Readiness	Classification
13/14	92.9%	Preliminary cross-site compatibility

Pillar Results

Category	MHDRM pillar	Response	Score	Flag
Data Characteristics	Modality Availability	1 — Partially demonstrated	1	
Data Characteristics	Data Quality and Completeness	2 — Demonstrated	2	
Data Characteristics	Population Representativeness	2 — Demonstrated	2	
Integration Readiness	Standardization and Interoperability	2 — Demonstrated	2	
Integration Readiness	Cross-Modal Alignment	2 — Demonstrated	2	
Organizational Readiness	Privacy and Security by Modality	2 — Demonstrated	2	
Organizational Readiness	Provenance and Governance	2 — Demonstrated	2	

Evidence and Rationale

MHDRM pillar	Evidence reviewed and scoring rationale
Modality Availability	Modality-availability requirements are documented. All required modalities are available and confirmed via site audits, but imaging and laboratory coverage vary materially by site based on clinical appropriateness.
Data Quality and Completeness	Common data-quality dimensions and acceptance thresholds are documented. A standardized outcome variable label has been established. Each site has completed an inventory of their local data and provided aggregated results for evaluation.

MHDRM pillar	Evidence reviewed and scoring rationale
Population Representativeness	A target population has been defined and documented, along with relevant subgroups. Demographic and clinical distributions were compared across sites, and the consortium documented sampling, weighting, and subgroup-monitoring strategies for identified differences.
Standardization and Interoperability	Interoperability standards are documented and versioned. Sites use DICOM for imaging and agreed-upon mappings for laboratory codes and units.
Cross-Modal Alignment	Patient, encounter, and diagnostic-window linkage rules are documented and shared to all sites.
Privacy and Security by Modality	The consortium has established regular security audits, and privacy mechanisms by modality have been standardized and implemented.
Provenance and Governance	The consortium has established data governance protocols that are well documented, with clear ownership and responsibilities.

Prototype limitation: This report provides a preliminary preimplementation screening of cross-site data compatibility. The scores and thresholds have not been empirically validated and do not certify that an MMFL initiative will be successful.